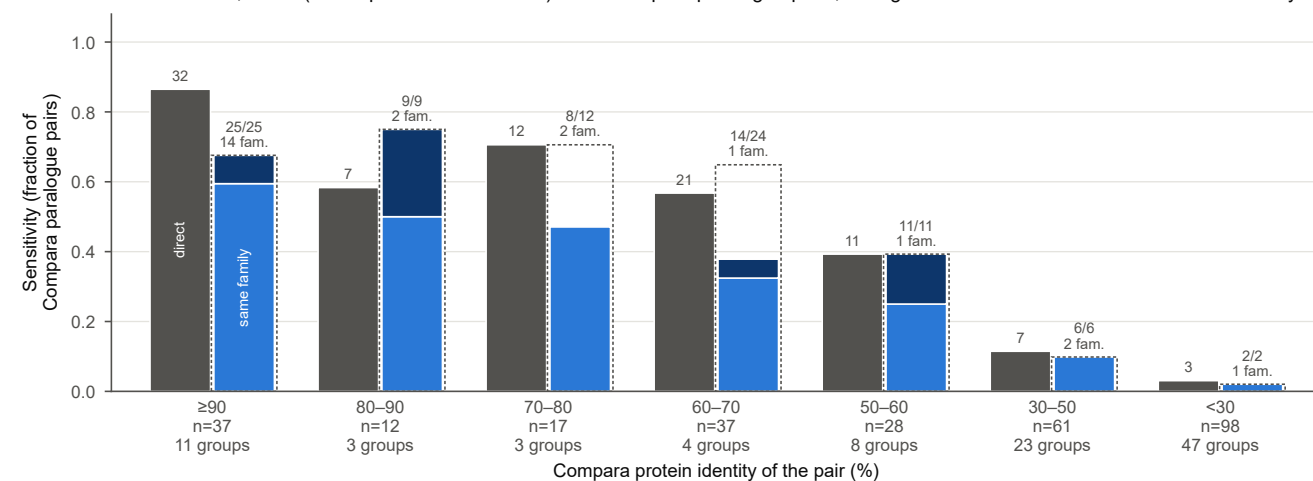


a Human A119b, chr16 (development chromosome) · 290 Compara paralogue pairs, both genes with a locus in Rustle's chr16 assembly



n = pairs in the band · groups = paralogue groups (connected components of the pairs) · dashed outline = both genes are family members (the most the families can recover)
k/m over the family bar = pairs in one family / pairs whose genes are both family members · fam. = default families holding the recovered pairs

b Human chr16 · 60–90%, by paralogue group

15 NPIP genes: one paralogue group, 59 pairs

Direct alignment between the genes' loci

40/59

Same default family

26 also aligned directly 5 31/59

6 other paralogue groups: 7 pairs

Direct alignment between the genes' loci

0/7

Same default family

0/7

c Human chr16 · precision over judgeable pairs

Direct alignment, nucleotide (asm20 -k11 -w5)

93/95

Same default family, all copies

75/88

from 16 families; 66 of the 88 pairs in one NPIP family; without it 28/30

Same default family, multi-exon copies only

71/83

from 14 families; 66 of the 83 pairs in one NPIP family; without it 18/19

Sensitivity (fraction of the group's pairs)

Precision against Compara (fraction of judgeable pairs)

Rustle's default de novo families: reads → loci (seeded with secondary alignments within 2% of the best score) → one representative per locus → families de novo (at least 80% genes with reciprocal alignments shared; Markov clustering). No protein